



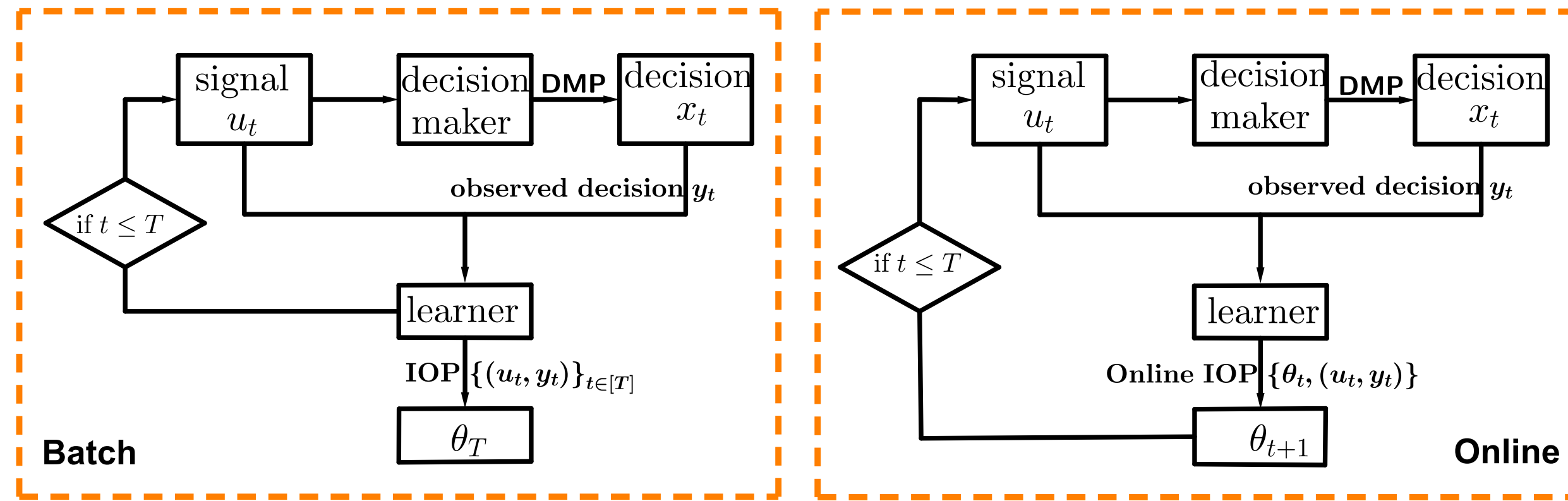
Generalized Inverse Optimization through Online Learning

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Introduction

- To unlock the potential of inverse optimization, we aim to elicit decision maker's preferences or restrictions through online learning in this paper.
- We develop an online learning algorithm to derive unknown parameter occurring in either the objective function or constraints.
- We also show that the online learning algorithm has a $\mathcal{O}(\sqrt{T})$ regret bound.



Decision making problem

We consider a family of parameterized optimization problem, in which $\mathbf{x} \in \mathbb{R}^n$ is the decision variable, $u \in \mathbb{R}^m$ is the external signal, and $\theta \in \mathbb{R}^p$ is the parameter.

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^n} \quad & f(\mathbf{x}, u, \theta) \\ \text{s.t.} \quad & \mathbf{g}(\mathbf{x}, u, \theta) \leq \mathbf{0} \end{aligned} \quad \text{DMP}$$

where $f: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \mapsto \mathbb{R}$ is a real-valued function, and $\mathbf{g}: \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \mapsto \mathbb{R}^q$ is a vector-valued function. We denote $X(u, \theta) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{g}(\mathbf{x}, u, \theta) \leq \mathbf{0}\}$ the feasible region of DMP. We let $S(u, \theta) = \arg \min\{f(\mathbf{x}, u, \theta) : \mathbf{x} \in X(u, \theta)\}$ be the optimal solution set of DMP.

Inverse optimization and online setting

Inverse optimization

Specifically, we consider a situation where a decision $\mathbf{y}_i$ (with respect to an external signal u_i) for $i \in [N]$ are observed and recorded, the IOP model can be formulated to infer the parameter θ in DMP by minimizing an empirical loss as in the following

$$\min_{\theta \in \Theta} \frac{1}{N} \sum_{i=1}^N l(\mathbf{y}_i, u_i, \theta) \quad (\text{Batch-IOP})$$

where θ denotes the parameter in DMP, and $l(\mathbf{y}_i, u_i, \theta)$ is a loss function that captures the discrepancy between the model inferred from data and the actual one.

Online setting

- In our online learning setting, the signal-noisy decision pair become available to the learner one by one. Hence, the learning algorithm produces $(\theta_1, \dots, \theta_{T+1})$.
- Let $l(\mathbf{y}_t, u_t, \theta_t)$ denote the loss the learning algorithm suffers when it tries to predict the t th decision given u_t based on $\{(u_1, \mathbf{y}_1), \dots, (u_{t-1}, \mathbf{y}_{t-1})\}$.
- The goal of the learner is to minimize the regret, which is the cumulative loss $\sum_{t \in [T]} l(\mathbf{y}_t, u_t, \theta_t)$ against the possible loss when the whole batch of decisions are available. Formally, the regret is defined as

$$R_T = \sum_{t \in [T]} l(\mathbf{y}_t, u_t, \theta_t) - \min_{\theta \in \Theta} \sum_{t \in [T]} l(\mathbf{y}_t, u_t, \theta).$$

Online learning algorithm

- Given a (signal, noisy decision) pair $(u, \mathbf{y})$ and a hypothesis θ , we set the loss function as the minimum (squared) distance between $\mathbf{y}$ and the optimal solution set $S(u, \theta)$ in the following.

$$l(\mathbf{y}, u, \theta) = \min_{\mathbf{x} \in S(u, \theta)} \|\mathbf{y} - \mathbf{x}\|_2^2. \quad \text{Loss Function}$$

- Once receiving the t th (signal, noisy decision) pair $(u_t, \mathbf{y}_t)$, θ_{t+1} can be obtained by solving the following optimization problem:

$$\theta_{t+1} = \arg \min_{\theta \in \Theta} \frac{1}{2} \|\theta - \theta_t\|_2^2 + \eta_t l(\mathbf{y}_t, u_t, \theta), \quad (1)$$

where η_t is the learning rate in round t , and $l(\mathbf{y}_t, u_t, \theta)$ is defined in (Loss Function).

Algorithm 1 Implicit Online Learning for Generalized Inverse Optimization

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1: Input: (signal, noisy decision) pairs  $\{(u_t, \mathbf{y}_t)\}_{t \in [T]}$ 
2: Initialization:  $\theta_1$  could be an arbitrary hypothesis of the parameter.
3: for  $t = 1$  to  $T$  do
4:   receive  $(u_t, \mathbf{y}_t)$ 
5:   suffer loss  $l(\mathbf{y}_t, u_t, \theta_t)$ 
6:   if  $l(\mathbf{y}_t, u_t, \theta_t) = 0$  then
7:      $\theta_{t+1} \leftarrow \theta_t$ 
8:   else
9:     set learning rate  $\eta_t \propto 1/\sqrt{t}$ 
10:    update  $\theta_{t+1} = \arg \min_{\theta \in \Theta} \frac{1}{2} \|\theta - \theta_t\|_2^2 + \eta_t l(\mathbf{y}_t, u_t, \theta)$  (solve (1))
11:   end if
12: end for
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Theoretical analysis

- Regret bound

Theorem 1. Under suitable conditions, choosing $\eta_t = \frac{D\lambda}{2\sqrt{2}(B+R)\kappa\sqrt{t}}$, we have

$$R_T \leq \frac{4\sqrt{2}(B+R)D\kappa}{\lambda} \sqrt{T}.$$

where λ and κ are related to the smoothness of the objective functions, B, R and D are related to the data, feasible region and parameter space, respectively.

- Consistency of the estimator

Theorem 2. Under suitable conditions, choosing $\eta_t = \frac{D\lambda}{2\sqrt{2}(B+R)\kappa\sqrt{t}}$, we have

$$\frac{1}{T} \sum_{t \in [T]} l(\mathbf{y}_t, u_t, \theta_t) \xrightarrow{p} \mathbb{E} [l(\mathbf{y}, u, \theta^*)]$$

as T approaches to infinity. Here, θ^* minimizes the true risk $\mathbb{E} [l(\mathbf{y}, u, \theta)]$.

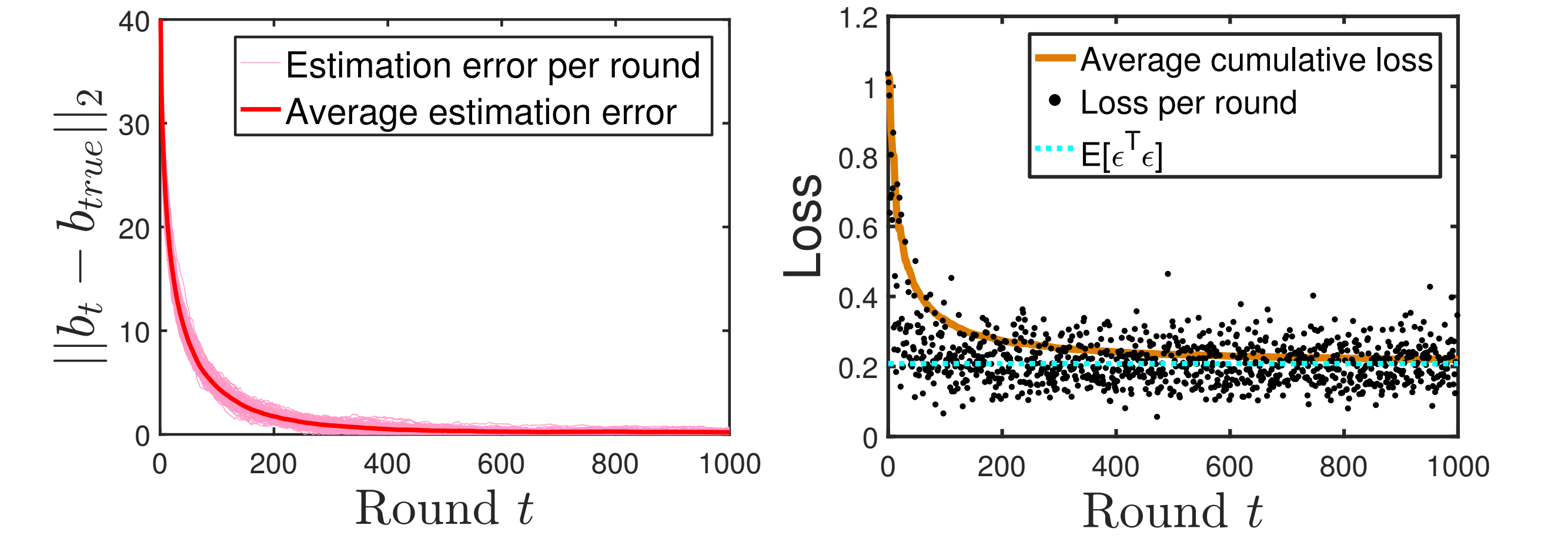
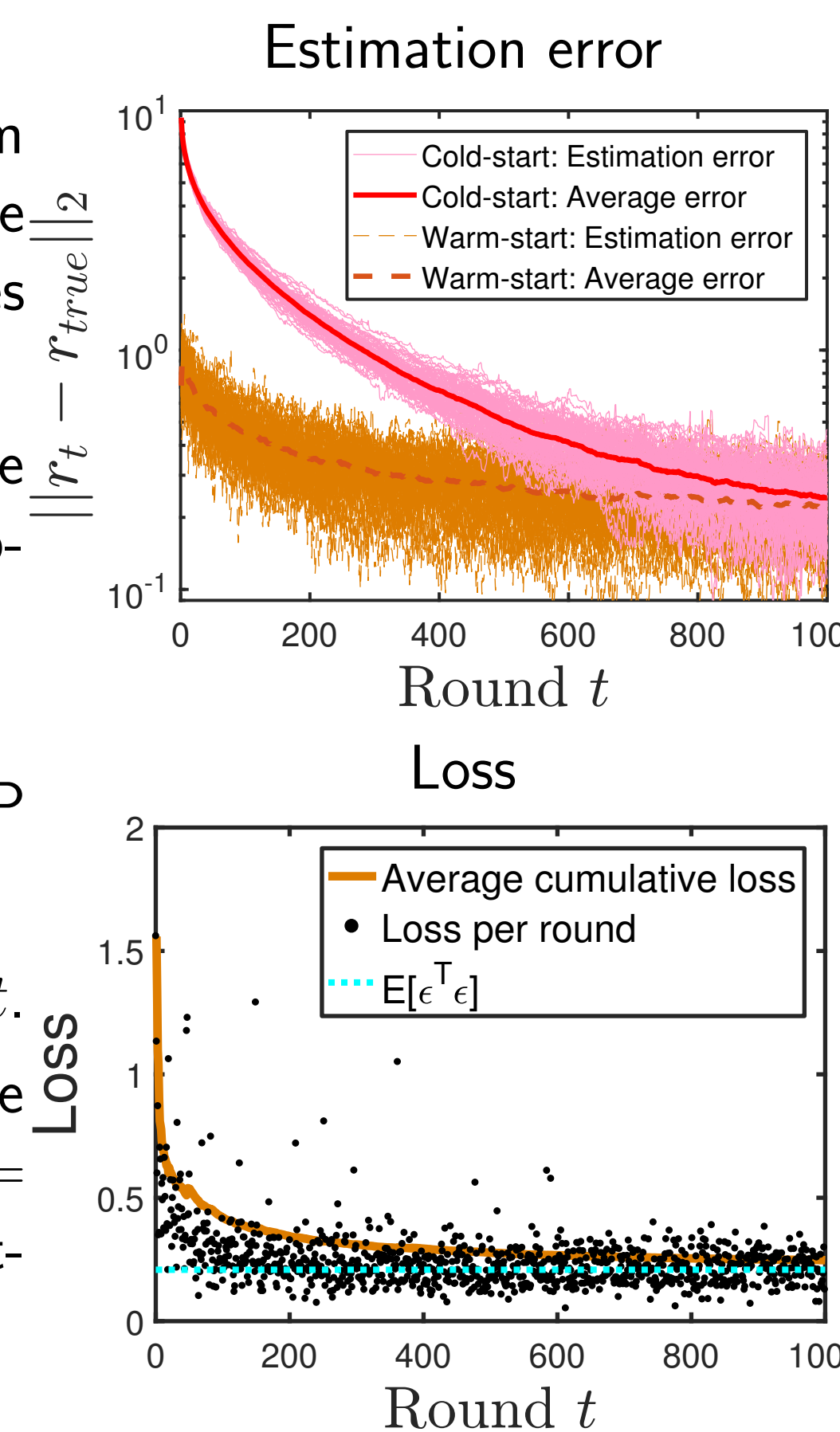
Learning consumer behavior

- We now study the consumer's behavior problem in a market with n products. The prices for the products are denoted by $\mathbf{p}_t \in \mathbb{R}_+^n$ which varies over different time $t \in [T]$.
- The consumer's decision making problem can be stated as the following utility maximization problem (UMP)

$$\begin{aligned} \max_{\mathbf{x} \in \mathbb{R}_+^n} \quad & u(\mathbf{x}) \\ \text{s.t.} \quad & \mathbf{p}_t^T \mathbf{x} \leq b \end{aligned} \quad \text{UMP}$$

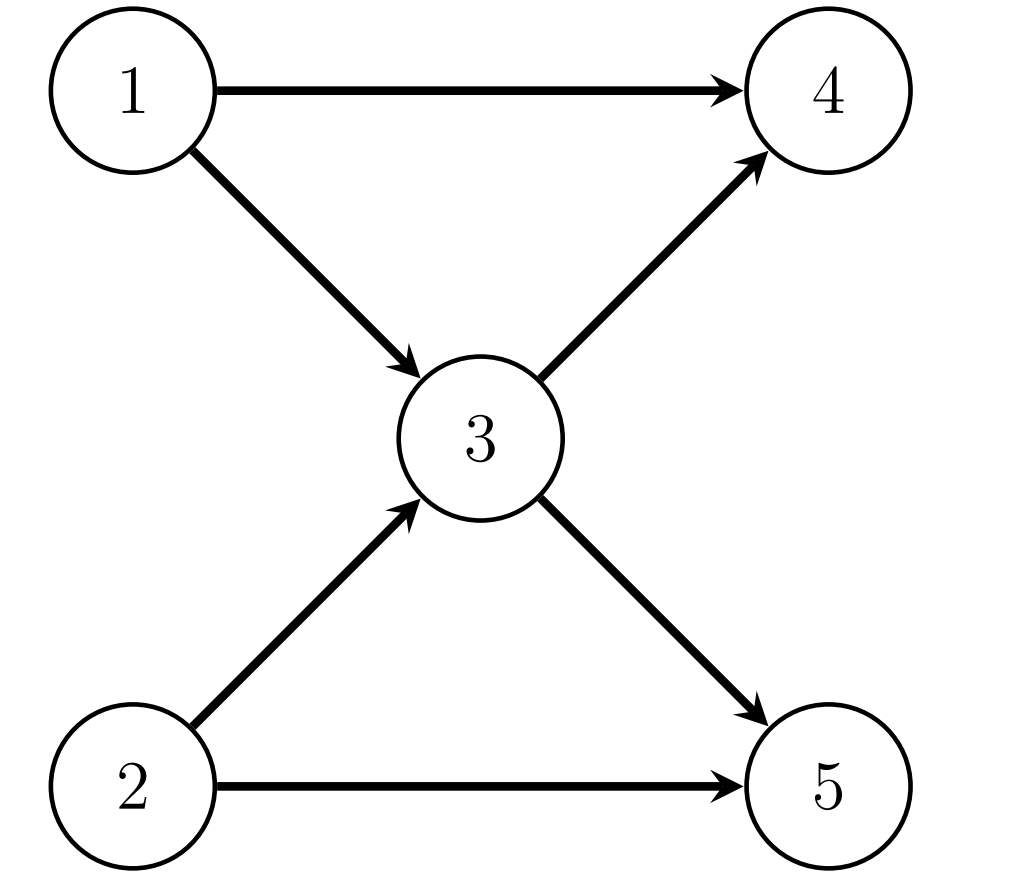
where $\mathbf{p}_t^T \mathbf{x} \leq b$ is the budget constraint at time t .

- In this application, we will consider a concave quadratic representation for $u(\mathbf{x})$. That is, $u(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T Q \mathbf{x} + \mathbf{r}^T \mathbf{x}$, where $Q \in S_-^n$ (the set of symmetric negative semidefinite matrices), $\mathbf{r} \in \mathbb{R}^n$.



Learning transportation cost

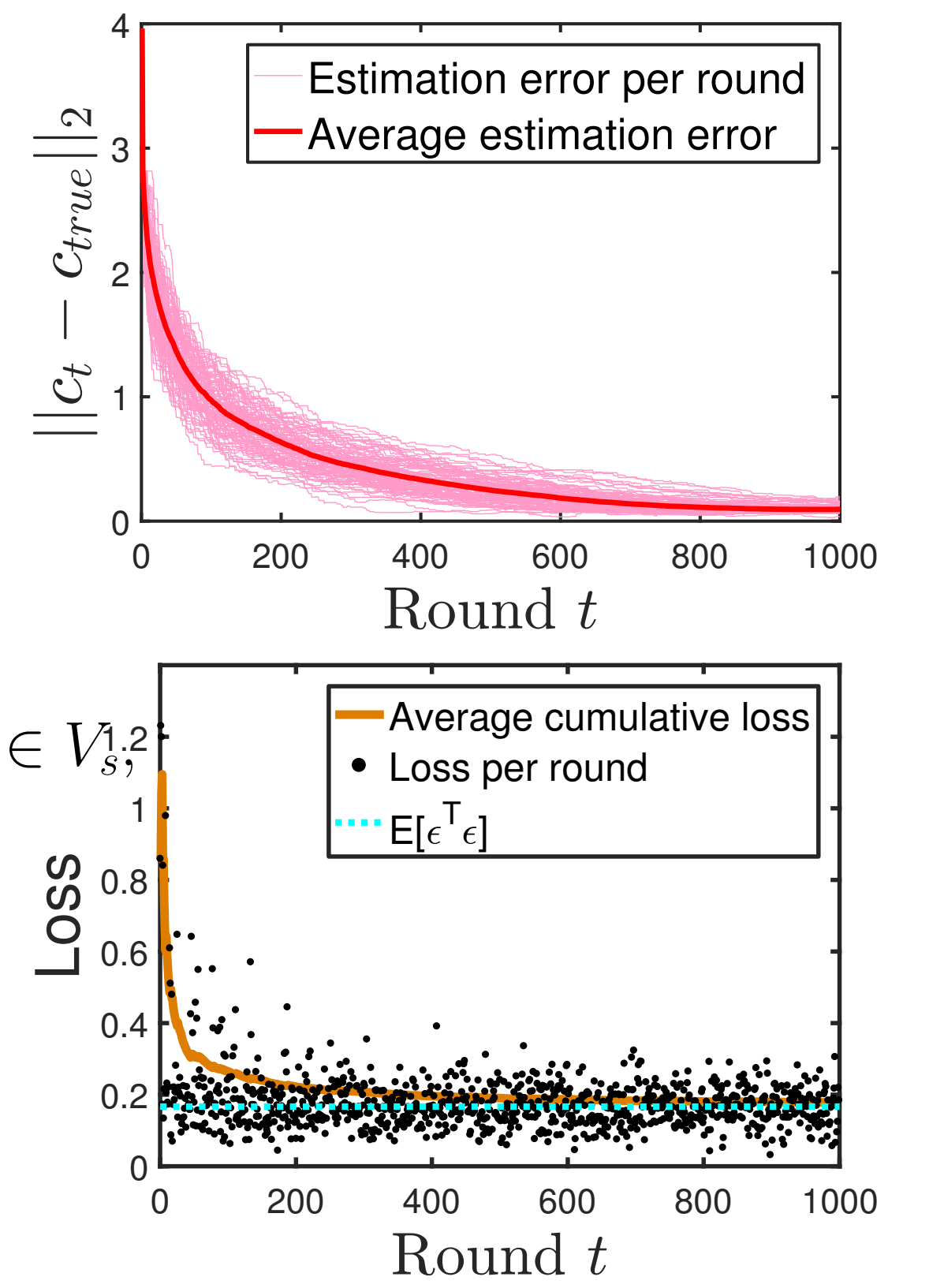
- We consider the transshipment network $G = (V_s \cup V_d, E)$, where nodes V_s are producers nodes V_d are consumers. The production level is y_v for node $v \in V_s$, and has a maximum capacity of w_v . The demand level is d_v^t for node $v \in V_d$ and varies over time $t \in [T]$.



- Let $C^v(y_v)$ be the producing cost for node $v \in V_s$. Let $c_e x_e$ be the transportation cost associated with edge $e \in E$, and the flow x_e has a maximum capacity of u_e .

- The transshipment problem is formulated as:

$$\begin{aligned} \min \quad & \sum_{v \in V_s} C^v(y_v) + \sum_{e \in E} c_e x_e \\ \text{s.t.} \quad & \sum_{e \in \delta^+(v)} x_e - \sum_{e \in \delta^-(v)} x_e = y_v, \quad \forall v \in V_s, \\ & \sum_{e \in \delta^+(v)} x_e - \sum_{e \in \delta^-(v)} x_e = d_v^t, \quad \forall v \in V_d, \\ & 0 \leq x_e \leq u_e, \quad 0 \leq y_v \leq w_v, \quad \forall e \in E, \forall v \in V_s, \end{aligned} \quad \text{TP}$$



where we want to learn the transportation cost c_e for $e \in E$. For this application, we will consider a convex quadratic cost for $C^v(y_v)$. That is, $C^v(y_v) = \frac{1}{2} \lambda_v y_v^2$, where $\lambda_v \geq 0$.

Conclusions

- We propose the first general framework for eliciting decision maker's preferences and restrictions using inverse optimization through online learning.
- This framework can learn general convex utility functions and restrictions with observed noisy signal-decision pairs.
- Moreover, we prove that the online learning algorithm, which uses an implicit update rule, has a $\mathcal{O}(\sqrt{T})$ regret under certain regularity conditions. Hence, this method has a fast convergence rate.

Code at <https://github.com/chaoshengdong/GIOPOL>
 Paper at <https://arxiv.org/pdf/1810.01920.pdf>

